

3 Methodology

We use a difference-in-differences design with continuous treatment intensity to estimate the effect of LLM exposure on UK labor market outcomes. Our identification strategy exploits variation in occupational and firm-level exposure to LLM capabilities, as measured by Eloundou et al. (2024), combined with the timing of ChatGPT’s public release in November 2022. We conduct three types of analyses: firm-level employment stocks using LinkedIn profile data, firm-level hiring decisions based on job posting data, and occupation-level hiring patterns using job posting data.

For our firm-level employment stock analysis, we estimate the following specification:

$$Y_{it} = \sum_{\tau=-15}^{30} \beta_{\tau} \cdot \text{Exposure}_i \times \mathbb{1}_{[t=\tau]} + \gamma_i + \delta_{jt} + \epsilon_{it} \quad (3)$$

where Y_{it} represents outcomes for firm i in month t , including log total employment, log employment by seniority level (junior vs. senior), the fraction of low-seniority positions, and average compensation. Exposure_i is the firm’s pre-treatment LLM exposure score, calculated as the employment-weighted average of occupational exposure scores using the workforce composition on 30 October. We standardize the exposure coefficients across all analyses to make the coefficients more easily interpretable. The term $\mathbb{1}_{[t=\tau]}$ denotes indicators for event time τ relative to November 2022, with October 2022 ($\tau = -1$) as the reference period. We include firm fixed effects γ_i to control for time-invariant firm characteristics, and industry-by-period fixed effects δ_{jt} to account for sector-specific shocks that might correlate with technology adoption patterns. Standard errors are clustered at the firm level.

This event study design allows us to test for parallel pre-treatment trends, as well as to examine the evolution of treatment effects over time, showing when and how labor market impacts start to emerge, and whether they persist, intensify, or attenuate over time. We use 15 months of pre-treatment data and 30 months of post-treatment data. To summarize the overall treatment effect, we also report average post-treatment effects from a static specification:

$$Y_{it} = \beta \cdot \text{Exposure}_i \times \text{Post}_t + \gamma_i + \delta_{jt} + \epsilon_{it} \quad (4)$$

where Post_t is an indicator for periods after November 2022.